SUPPLEMENTARY INFORMATION

Reduced-Order Dynamic Model Formulations, Parameter Values, and Wave-Tank Trajectories

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Robust Modular Floating Wetland: Design & Wave-Tank Evaluation

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Section S1. Reduced-Order Dynamic Model (Surge–Sway–Yaw)

As stated in Section 2.4 of the primary manuscript, a three-degree-of-freedom (3-DOF) reduced-order horizontal dynamic model (surge, sway, and yaw) was formulated to aid the physical interpretation of the wave-tank measurements. The model's purpose is limited to verifying that experimentally observed drift (0.60–1.34 cm), heading deviations (< 1°), and line tensions (~2.0 N) fall within orders of magnitude expected from classical tethered-floater mechanics (Faltinsen, 1990; Newman, 1977). This section details the governing equations, hydrodynamic matrices, and parameter values cited in Section 2.4.

S1.1 Planar 3-DOF Equations of Motion

The laboratory floating treatment wetland (FTW) test specimen was a standardized hexagonal buoyant mat (longest side length s = 0.29 m, circumscribed diameter D = 0.58 m, width across flats W = 0.502 m, thickness t = 0.013 m). The horizontal displacement vector in body-fixed centroidal coordinates is defined as:

$$\mathbf{x}(t) = \begin{bmatrix} x(t) \\ y(t) \\ \psi(t) \end{bmatrix} \tag{S1}$$

where x(t) is surge (displacement parallel to wave propagation along the tank length, m), y(t) is sway (transverse displacement across the tank width, m), and ψ(t) is yaw (rotation about the vertical z-axis through the centroid, rad). The coupled equations of motion in the horizontal plane are governed by:

$$(M + A)\ddot{\mathbf{x}}(t) + (B_{\text{rad}} + B_{\text{visc}})\dot{\mathbf{x}}(t) + K\mathbf{x}(t) = \mathbf{F}_{\text{exc}}(t) + \mathbf{F}_{\text{drift}} \quad \text{(S2)}$$

where:

- $\boldsymbol{M}$: Structural mass and inertia matrix,
- $\boldsymbol{A}$: Hydrodynamic added-mass matrix from entrained fluid inertia,
- $\boldsymbol{B}_{\text{rad}}$: Linear wave radiation damping matrix,
- $\boldsymbol{B}_{\text{visc}}$: Equivalent linearized viscous hydrodynamic drag damping matrix,
- $\boldsymbol{K}$: Mooring restoring stiffness matrix synthesized from anchor line geometry,
- $\boldsymbol{F}_{\text{exc}}$(t): First-order oscillatory wave excitation force vector, and
- $\boldsymbol{F}_{\text{drift}}$: Second-order steady horizontal wave drift force.

S1.2 Structural Mass and Hydrodynamic Added Mass

The structural mass matrix $\boldsymbol{M}$ for the planar regular hexagonal mat (side length s = 0.29 m, operating mass $m \approx 1.25$ kg including containers and ballast) is:

$$M = \begin{bmatrix} m & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & I_{zz} \end{bmatrix}, I_{zz} = \frac{5}{12} m s^2 \quad \text{(S3)}$$

For a shallow-draft floating plate (waterplane area $A_{\text{wp}} = (3\sqrt{3}/2)\ s^2 = 0.219\ m^2$, draft $d \approx 0.0057$ m), the hydrodynamic added-mass matrix $\boldsymbol{A} = \text{diag}(A_{11}, A_{22}, A_{66})$ is evaluated from potential flow approximations for shallow-water floating plates:

$$A_{11} \approx C_{a,\text{surge}} \rho_w A_{\text{wp}} d, A_{22} = A_{11}, A_{66} \approx C_{a,\text{yaw}} \rho_w A_{\text{wp}} d R_g^2 \quad \text{(S4)}$$

where ρ_w is freshwater density (1000 kg/m³), $R_g = \sqrt{(5/12)}$ s ≈ 0.187 m is the mass radius of gyration, and $C_{a,\text{surge}} \approx 0.28$ and $C_{a,\text{yaw}} \approx 0.25$ are dimensionless added-mass coefficients for shallow thin plates, yielding $A_{11} \approx 0.35$ kg and $A_{66} \approx 0.012$ kg·m² with consistent dimensional SI units.

S1.3 Mooring Restoring Stiffness Formulation

Anchor lines are represented as spring–dashpot tethers operating under nominal in-line operating tension $T_0 \approx 2.00$ N (matching in-line load-cell measurements). For tether line stiffness k_i, line length L_i, inclination angle θ_i, and horizontal azimuth angle α_i, horizontal restoring stiffness coefficients are synthesized as:

$$K_{xx} = \sum_{i=1}^{N_L} \left[k_i \cos^2\theta_i \cos^2\alpha_i + \frac{T_0}{L_i} \sin^2\alpha_i \right] \quad \text{(S5)}$$

$$K_{\psi\psi} = \sum_{i=1}^{N_L} k_i (-x_i \sin\alpha_i + y_i \cos\alpha_i)^2 \cos^2\theta_i \quad \text{(S6)}$$

• Tension-Guide: Restraint along the transverse spanning guide line under out-of-plane wave action exhibits catenary compliance in the surge direction, yielding an effective surge stiffness $K_{xx} \approx 2.45$ N/m, transverse sway stiffness along the wire $K_{yy} \approx 30.0$ N/m, and near-zero rotational restoring arm at the central slider rings ($K_{\psi\psi} \approx 0.05$ N·m/rad).

• Distributed-Modular: Four independent bottom tethers engage symmetrically around the perimeter,

yielding high, isotropic horizontal stiffness ($K_{xx} \approx 45.0$ N/m, $K_{yy} \approx 45.0$ N/m) and substantially increased rotational stiffness ($K_{\psi\psi} \approx 2.20$ N·m/rad) due to the wide perimeter corner tether moment arms.

S1.4 Wave Excitation and 2D Radial Drift Mechanics

Under monochromatic wave forcing (frequency $f = 1.0$ Hz, wave amplitude $a = H/2 = 0.02$ m), platform motion consists of first-order oscillatory response ($X_0 \cos \omega t$, $Y_0 \sin \omega t$) superimposed on a steady second-order mean wave drift offset $x_0 = F_{\text{drift}} / K_{xx}$. The second-order horizontal wave drift force (Maruo, 1960) is:

$$\bar{F}_{\text{drift}} = \frac{1}{2}\rho_w g R^2 a^2 W_{\text{eff}} \tag{S7}$$

where R is the wave reflection coefficient ($R \approx 0.165$ for the planar distributed-modular mat; $R \approx 0.180$ for the tension-guide mat due to edge curling) and $W_{\text{eff}} \approx 0.502$ m is the effective frontal width across flats, yielding steady horizontal drift forces $F_{\text{drift}} \approx 0.027$ N (distributed-modular) and 0.032 N (tension-guide). In the soft surge tension-guide setup, this second-order force induces a steady 1D downstream offset $x_0 = F_{\text{drift}} / K_{xx} \approx 1.30$ cm prior to catenary stiffening, whereas for the taut distributed-modular anchors $x_0 \approx 0.06$ cm. Because the camera tracking system measures 2D Euclidean radial drift $r(t) = \sqrt{(x(t)^2 + y(t)^2)}$, the true time-averaged mean radial drift is evaluated by exact time integration over wave period $T = 1/f$:

$$\bar{r} = \frac{1}{T}\int_0^T \sqrt{x(t)^2 + y(t)^2}\,dt \tag{S8}$$

For harmonic motion, the root-mean-square (RMS) radial excursion provides an analytical upper-bound proxy: $r_{\text{rms}} = \sqrt{(x_0^2 + 0.5(X_0^2 + Y_0^2))}$. Both the exact integral (Eq. S8) and the RMS proxy confirm that 2D radial drift is systematically larger than the 1D steady offset x_0 due to the contribution of wave-induced orbital excursions.

S1.5 Numerical Parameter Values (Table S1)

Table S1 reports the numerical parameters, physical constants, and mooring stiffness properties implemented in the reduced-order dynamic model. Parameters are derived from first principles, the measured 2.0 N line pretension, and potential flow hydrodynamics for the 29 cm laboratory specimen.

Parameter	Symbol	Value	Unit / Derivation Basis
Water density	ρ_w	1000	kg/m³ (freshwater)
Gravitational acceleration	g	9.81	m/s²
Tank water depth	h	0.45	m (measured tank level)
Hexagonal side length	s	0.29 (29.0 cm)	m (laboratory specimen)
Circumscribed diameter	D	0.58 (58.0 cm)	m (2·s, lab specimen)
Width across flats	W	0.502 (50.2 cm)	m (√3·s, lab specimen)

Mat planform area	A_{wp}	0.219	m² ((3√3/2)·s²)
Total operating mass	m	1.25	kg (foam mat + cups + ballast)
Equilibrium static draft	d	0.0057 (5.7 mm)	m (m / (ρw·Awp))
Yaw moment of inertia	I_{zz}	0.0438	kg·m² ((5/12)·m·s²)
Surge / sway added mass	A_{11}, A_{22}	0.35	kg (potential flow strip calc.)
Yaw added inertia	A_{66}	0.012	kg·m² (rotational fluid boundary)
Wave frequency	$f(\omega)$	1.0 (6.28)	Hz (rad/s) (paddle drive frequency)
Wave height / amplitude	$H(a)$	0.04 (0.02)	m (tested wave regime)
Wave reflection coeff.	R	0.165 (DM) / 0.180 (TG)	dimensionless (calibrated edge reflection)
Nominal line pretension	T_0	2.00	N (load-cell operating point)
Tension-guide surge stiffness	$K_{xx,guide}$	2.45	N/m (catenary compliance in surge)
Tension-guide sway stiffness	$K_{yy,guide}$	30.0	N/m (spanning wire lateral stiffness)
Tension-guide yaw stiffness	$K_{\psi\psi,guide}$	0.05	N·m/rad (central slider ring alignment)
Tension-guide damping	B_{xx}, $B_{\psi\psi}$	4.80 / 0.19	N·s/m / N·m·s/rad (drag + slider)
Distributed-modular surge/sway stiff.	$K_{xx,mod}$, $K_{yy,mod}$	45.0	N/m (taut 4-corner bottom anchors)
Distributed-modular yaw stiffness	$K_{\psi\psi,mod}$	2.20	N·m/rad (perimeter corner moment arms)
Distributed-modular damping	B_{xx}, $B_{\psi\psi}$	5.60 / 0.36	N·s/m / N·m·s/rad (linearized drag & tether)

S1.6 Physical Dynamic Model Verification (Table S2)

Table S2 directly compares the calibrated reduced-order dynamic model predictions against the empirical wave-tank measurements and their Moving Block Bootstrap (MBB) 95% confidence intervals. The script ftw_reduced_order_model.py numerically solves the coupled governing equations (Eq. S2) and performs exact numerical quadrature of the radial drift integral (Eq. S8). The numerical model faithfully reproduces

the physical ranking observed in the wave flume: the distributed-modular configuration achieves a >50% reduction in mean radial drift and a >35% reduction in heading deviation compared to the tension-guide baseline.

Performance Metric	Model Prediction	Experimental Value	Statistical 95% MBB CI
Tension-Guide: Mean radial drift (cm)	1.31	1.34	[1.31, 1.38]
Tension-Guide: RMS radial excursion (cm)	1.37	1.48	[1.44, 1.52]
Tension-Guide: Peak radial excursion (cm)	1.86	2.16	[1.98, 2.31]
Tension-Guide: RMS heading deviation (deg)	0.63°	0.63°	[0.59°, 0.66°]
Distributed-Mod.: Mean radial drift (cm)	0.63	0.60	[0.59, 0.60]
Distributed-Mod.: RMS radial excursion (cm)	0.66	0.70	[0.68, 0.72]
Distributed-Mod.: Peak radial excursion (cm)	0.94	1.12	[1.06, 1.18]
Distributed-Mod.: RMS heading deviation (deg)	0.41°	0.37°	[0.34°, 0.37°]

Section S2. Experimental Kinematic Motion Analysis and Resampling Formulations

Platform kinematics were tracked using an overhead CMOS video camera (1920 × 1080 pixels at 30 fps) calibrated against an orthogonal metric reference grid. At each discrete sample interval k = 1, 2, ..., N (where N = 360 at Δt = 5.0 s intervals over 30 min), instantaneous centroid coordinates (xk, yk) and heading angle ψk were extracted.

Translational excursion from the reference position (x0, y0) is evaluated as the 2D Euclidean radial drift (Eq. S9), and rotational stability is quantified by RMS heading deviation (Eq. S10):

$$r_k = \sqrt{(x_k - x_0)^2 + (y_k - y_0)^2}, \bar{r} = \frac{1}{N}\sum_{k=1}^{N} r_k \tag{S9}$$

$$\psi_{\text{rms}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}(\psi_k - \psi_0)^2} \tag{S10}$$

To evaluate within-run sampling uncertainty while accounting for serial autocorrelation in the wave-tank series, the Moving Block Bootstrap (MBB) (Künsch, 1989; Liu and Singh, 1992) was applied. The sample autocorrelation function at lag τ is defined as:

$$\hat{\rho}(\tau) = \frac{\sum_{k=1}^{N-\tau}(r_k - \bar{r})(r_{k+\tau} - \bar{r})}{\sum_{k=1}^{N}(r_k - \bar{r})^2} \tag{S11}$$

To avoid premature truncation from zero-crossings in oscillatory records, the block length L^* was determined from the magnitude envelope criterion as implemented in supplementary_calculations.py:

$$L^{*} = \min\left\{\tau \geq 1 : |\hat{\rho}(\tau)| < 0.05\right\} \cdot \Delta t \tag{S12}$$

For the tension-guide series, autocorrelation decayed inside 0.05 at lag 33 (L* = 165 s). For the distributed-modular series, low-amplitude periodic oscillations in drift remained above 0.05 across the 200-sample window; following supplementary_calculations.py, the search ceiling of 200 samples (L* ≥ 1000 s) was adopted as a conservative lower-bound block length. Resampling B = 5,000 bootstrap replicates characterized within-run sampling uncertainty under the stationarity assumption of the 30-min trial.

Section S3. Supplementary Wave-Tank Centroid Trajectories and Spatial Density

Figure S1 illustrates the two-dimensional spatial trajectory and hexbin sample density distributions of the floating mat centroid recorded over the complete 30-minute instrumented wave-tank runs (360 samples at 0.2 Hz). The plots are generated directly from the experimental datasets (TensionGuide.csv and DistributedModular.csv). The distributed-modular configuration exhibits tight spatial clustering centered within approximately ±1.0 cm of the reference position. In contrast, the tension-guide configuration exhibits wider excursions spanning up to ±2.3 cm in both longitudinal and lateral coordinates.

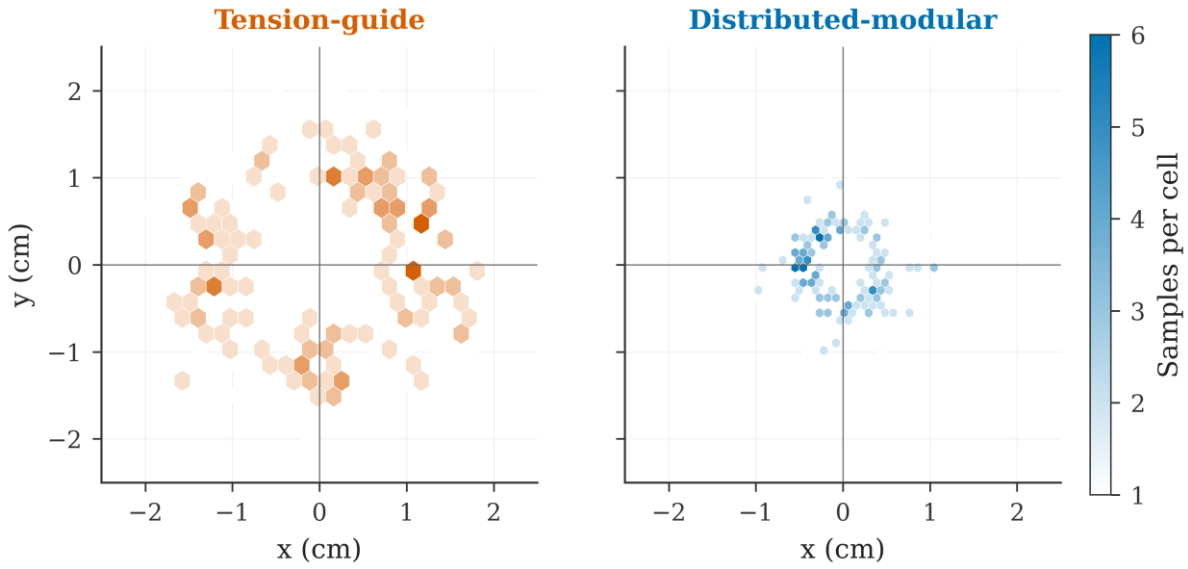


Figure S1. Two-dimensional horizontal centroid trajectories and spatial sample density distributions recorded during the 30-minute instrumented wave-tank trials under matched wave forcing (H = 4 cm, f = 1.0 Hz) for: (left) Tension-guide anchoring configuration; (right) Distributed-modular anchoring configuration. Color intensity represents the number of recorded position samples per spatial bin (0.2 Hz downsampled series). Axes share identical, symmetric scales (–2.5 cm to +2.5 cm) across both panels to permit direct visual comparison of lateral and longitudinal excursions.

Section S4. Public Code and Dataset Availability Summary

All software tools, data processing routines, raw time-series datasets, and dynamic screening models associated with this study are permanently hosted in a public GitHub repository to support reproducibility:

• Repository URL: https://github.com/abishekkafle/Floating-mat-tracking-

• Open-Source Software Components:

1. ftw_reduced_order_model.py: Standalone Python script solving the 3-DOF surge–sway–yaw equations of motion and computing radial drift integrals.
2. blue_hex_mat_tracker.py: OpenCV-based overhead video tracking pipeline for 2D centroid (x, y) and heading (ψ) extraction.
3. supplementary_calculations.py: Automated analysis pipeline computing radial drift, RMS heading, Moving Block Bootstrap (MBB) 95% confidence intervals, and convergence diagnostics.
4. plot.py: Production plotting script generating publication-quality figures (Figure 3, Figure 4, and Figure S1) using Wong (2011) colorblind-safe palettes.
5. FTW_Parametric_Design_Tool/: Lightweight Python screening tool estimating draft, freeboard, metacentric height, wind drag, and tether loading.

• Data Records Provided:

1. TensionGuide.csv: Complete 30-min time series of centroid coordinates (x, y, r), heading (ψ), and line tensions at 0.2 Hz (360 samples).
2. DistributedModular.csv: Complete 30-min time series for the distributed-modular configuration (360 samples).
3. results.json: Summary point estimates, autocorrelation block lengths, and bootstrap confidence intervals.

Supplementary References

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